

Repressive Democracies: State Repression, Protest, and Policy Change

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Abstract

This paper studies how the cost of preparing for mass protest affects a democratic state's choice of repression. Citizens coordinate under uncertainty about the participation threshold required for policy change. The paper adopts the psychological-rewards global game of Bueno de Mesquita and Shadmehr (2023) as its participation stage. It adds a state that values the policy status quo, plans operational resources for expected protest volume, and bears a convex political cost from authorizing expected harm per participant. The central result is that a higher marginal cost of preparing for protesters induces greater expected participant harm when the resources saved through deterrence exceed the additional operational cost of harsher tactics. Two police technologies show why the effect is positive in deployment-intensive systems and can be negative when the same inputs are important for implementing force. Democratic accountability enters through the marginal political-cost schedule. Stronger constraints reduce the state's chosen policy. A concise extension allows repression to change protest motivation. Repression backfires precisely when the elasticity of success-contingent motivation to expected harm exceeds one.

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1 Introduction.

Studies disagree about how state repression affects mobilization (Davenport 2007a). Some find that threats of repression deter mobilization by raising the cost of protest (Chenoweth, Perkoski, and Kang 2017; della Porta 1995; Christopher M. Sullivan 2016). Others find that repression can backfire by radicalizing victims of state violence (Adam-Troian, Çelebi, and Mahfud 2020; Christopher M. Sullivan and Davenport 2017). Repression and dissent become strategically interdependent when states choose the severity of repression (Chenoweth, Perkoski, and Kang 2017; Ritter 2014).

Scholars call this phenomenon the “protest-repression” nexus (Davenport 2007a; Curtice and Behlendorf 2021), and they have studied it primarily in autocracies. An empirical strand examines protest in democracies (Carey 2006; Nam 2007; Machado, Scartascini, and Tommasi 2011; Su 2015; Aytaç, Schiumerini, and Stokes 2017). It asks mainly what makes protest more or less likely across democratic settings. The literature has paid less attention to how severely states choose to repress protest. Competitive elections, protected speech, and organized civil society make protest a viable complement to representative channels, while legal and electoral accountability make the costs of coercion especially visible. Democratic protest policing therefore places deterrence, operational capacity, and political accountability in the same strategic problem.

Mass protests are costly for incumbent elites in democracies for two reasons. A massive protest may force politicians to change policies they prefer to preserve (Bosi, Giugni, and Uba 2016; Klein and Regan 2018). Gridlocked constitutional provisions that benefit elites are one example, including the tax code or the electoral system. Massive protests also require the state to deploy personnel, equipment, transport, detention capacity, and other operational resources.

To study this strategic problem, I develop a sequential model of policy-change protest. The elected executive first sets the expected repression cost borne by a participant. A continuum of citizens then decides whether to protest. The protest changes policy only if participation reaches an uncertain critical threshold, and each citizen observes a noisy private signal of that threshold. Participants receive a success-contingent expressive reward and bear the expected repression cost. The state values preservation of the disputed status quo. It plans operational resources for the expected volume of protest and bears a convex political cost from expected participant harm. This

structure places mobilization, deterrence, and the state's two distinct costs in a single strategic system.

Democratic institutions determine how likely a participant is to be victimized and how much physical or psychological damage state agents may inflict. These institutions include legal rules, standing procedures, and political oversight. Within that framework, elected executives and police commanders set deployments, authorized tactics, and rules of engagement for anticipated protest episodes. The model represents these choices by the expected repression cost announced or credibly anticipated before mobilization and fixed while an episode unfolds. Political leaders weigh its deterrent value against legal, electoral, administrative, and reputational consequences (Nordås and Davenport 2013; Ritter 2014).

The paper isolates one mechanism. A costly crowd gives the state an incentive to deter participation, but harsher policy creates political costs and can require additional operational resources. I represent repression by the expected cost borne by a participant. This measure combines the reach and intensity of coercion. The state's optimal policy balances the status quo it protects and the planned resources it saves by reducing expected protest volume against the political and operational costs of harsher tactics. I identify the precise technological condition under which more expensive protest preparation produces harsher repression.

The participation benchmark is inherited from Bueno de Mesquita and Shadmehr's (2023) psychological-rewards setting. A continuum of nonpivotal citizens receives private signals about the critical participation threshold, and a success-contingent reward generates strategic complementarity. The present paper changes the government's problem. Its payoff distinguishes the political value of the disputed status quo, the planned operational cost of expected protest volume, and the political cost of expected participant harm.

Table 1 identifies the exact analytic elements retained from the benchmark and the changes that generate this paper's contribution.

Table 1: Inherited benchmark and this paper’s contribution

Analytic element	Inherited from Bueno de Mesquita and Shadmehr (2023)	Treatment in this paper
Citizen stage	A continuum receives noisy private signals about an unknown participation threshold; success is determined by a monotone cutoff equilibrium	<i>Retained.</i> The same coordination game is applied to democratic protest for policy change
Marginal incentive	A nonpivotal citizen compares a success-contingent psychological reward with the individual cost of repression	<i>Reparameterized.</i> d_p is participation-contingent motivation; $c_r = q_r s_r$ is expected harm per participant
State timing	The state chooses repression before citizens mobilize	<i>Institutionally specified.</i> The executive chooses after the policy dispute is known and holds the directive fixed during mobilization
State payoff	Repression deters participation and carries a direct convex cost	<i>New payoff.</i> It separates the status quo stake B_s , planned operational cost $H(\bar{N}, c_r; \theta_0)$, and political cost $K(c_r; \theta_1)$
Central result	Optimal repression balances deterrence against the direct cost of repressive capacity	<i>New result.</i> A planning-cost shifter raises c_r^* exactly when the resources saved through deterrence exceed its direct effect on the cost of harsher tactics
Motivation extension	Success-contingent motivation is fixed	<i>New extension.</i> Repression may change motivation; backlash occurs when its elasticity to c_r exceeds one

The rest of the paper reviews the relevant protest-repression and coordination literatures, presents the inherited participation benchmark, develops the state’s crowd-cost mechanism and its comparative statics, and discusses the paper’s theoretical and empirical contribution.

2 Literature review.

2.1 State repression, political attitudes, and social movements.

Research on state repression examines why governments coerce citizens and how coercion affects political behavior, especially protest. Political repression is the use of institutional or physical force, most commonly violent, to subdue or coerce individuals and groups that challenge political actors or the status quo (deMeritt 2016). Much of this research focuses on authoritarian regimes or flawed democracies, where rulers face weak accountability and citizens lack protected speech

or competitive elections.

Comparative research has persistently found that democratic countries exhibit lower levels of state repression (Davenport 2007a, 2007b; Davenport and Armstrong 2004). Yet violent coercion by state agents also occurs in developed democracies (Belur 2010; Whitehead, Fair, and Payne 2010; Worden 1995). These episodes affect public opinion, as support for the police fell sharply after police killed George Floyd (Reny and Newman 2021). Victims of police repression may also experience anxiety, depression, anger, and fear (Bor et al. 2018; Çelebi, Adam-Troian, and Mahfud 2020; Cooper et al. 2004; J. E. DeVolder et al. 2017; J. DeVolder, Fedina, and Link 2020).

Yet not all citizens perceive the use of force by repressive bureaus the same way. In the United States, for example, studies consistently find that conservatives support the police at higher rates than liberals (Ekins 2016; Fingerhut 2017; Hindelang 1974; Zamble and Annesley 1987). Voters may perceive and evaluate state repression differently, to the extent that they have directional motivations and are affective partisans. Attitudes about the police are also racially divided (Ekins 2016; Fingerhut 2017).

Davenport (2007a) calls the conflicting evidence about how repression affects social movements the “punishment puzzle.” Some studies find that repression discourages mass and peaceful protest but encourages radicalization (della Porta 1995). Others find that it increases both protest escalation and mobilization (Adam-Troian, Çelebi, and Mahfud 2020; Christopher M. Sullivan and Davenport 2017).

2.2 Empirical regularities and theoretical foundations.

The literature has identified several determinants that systematically explain state repression, among them regime type, economic development, population size, intrastate civil conflicts, and the degree of grievances and dissent (Davenport 2007b; Nordås and Davenport 2013; Christopher M. Sullivan 2016; Christopher M. Sullivan and Davenport 2017). For example, Nordås and Davenport (2013) argue that the demographic structure of the population is a key determinant of state repression. States respond with more repression where youth bulges are larger, meaning cohorts aged 15 to 24 relative to the overall adult population. States that anticipate more dissent raise repression preemptively, to deter it.

Chenoweth, Perkoski, and Kang (2017) summarize several consistent findings in the literature.

First, dissent precedes state repression in most settings, a regularity Davenport (2007a) calls the “law of coercive responsiveness.” Second, democratic countries tend to exhibit lower levels of repression. The authors also discuss the “punishment puzzle” mentioned above. They stress that repression and dissent, whether protest or other tactics, are two complex interlinked processes.

Formal models explain the relationship between state repression, protests, and regime or policy change in several ways. No definite consensus has emerged about the best modeling approach. Ritter (2014) notes that models choose different objects of investigation, and that each one, taken alone, misses important strategic features of the relation between repression and dissent. For example, some studies model the determinants of regime change or dissent in general as a process of an ongoing or expected conflict. In one approach, citizens protest to alter the status quo, and their success depends on the state’s concessions. A revolutionary vanguard’s level of violence may signal the level of dissent in society, which provokes citizens to cooperate (Bueno de Mesquita 2010). Other formal models turn to uncertainty and asymmetric information about government types (Pierskalla 2010). A “strong” state may repress citizens at high cost to signal its resolve. Citizens protest over repeated play and learn what type of government they face. Closest to the present paper in timing, De Jaegher and Hoyer (2019) model *preemptive* repression. A government sets repressive capacity before protest occurs, and dissidents respond by investing in countermeasures. De Jaegher and Hoyer derive both sides of the punishment puzzle. Repression deters when the benefit of protest is large relative to the cost of countermeasures, and backfires when it is small. Their complete-information coordination game has two dissidents, each of whom can determine the outcome. The relative size of the equilibria’s basins of attraction resolves the resulting multiplicity.

Thus, as Ritter (2014) notes, some models study the conditions that ignite dissent, and others study how the state chooses the severity of repression. But studies miss an important part of the puzzle when they treat these two processes separately. She therefore proposes a model that considers the conditions sparking dissent together with the state’s choice of severity. She models the strategic setting as a bargaining problem over policy provision and its effect on incumbent survival. Ritter finds that a higher likelihood of leader survival decreases the level of dissent and repression. Yet the same increase raises the severity of both.

2.3 Global games of protest and regime change.

This paper borrows its analytic setup from the global games literature on coordination under incomplete information (Carlsson and van Damme 1993; Morris and Shin 1998, 2003). The canonical regime change game has a continuum of citizens who simultaneously decide whether to attack a regime. The regime collapses if and only if the mass of attackers exceeds its strength. That strength is an uncertain fundamental, and citizens receive noisy private signals about it. Under complete information about the fundamental, the game exhibits multiple equilibria in an intermediate range of regime strength. The incomplete information selects a unique equilibrium in threshold strategies.

Two recent applications inform the present model directly. Bueno de Mesquita and Shadmehr (2023) develop “a standard global game of regime change.” A continuum of citizens receive noisy private signals $x_i = \theta + \sigma \varepsilon_i$ about the regime’s strength θ , and the rebellion succeeds if and only if the fraction of rebels exceeds θ . Their psychological-rewards setting yields the threshold $1 - c$, where c is the participation cost, and their government chooses costly repressive capacity before citizens move. The present paper takes that participation stage as its benchmark. Its contribution begins with the government’s payoff: the state values a particular policy status quo and separately accounts for the operational resources planned for expected protest volume and the political cost of expected participant harm. This distinction produces the crowd-management mechanism studied below.

The success-contingent reward also connects the model to work on selective incentives and democratic participation. Aytaç and Stokes (2019) explain participation through costs that citizens associate with abstention, including emotional responses and social obligations. Wood’s (2003) pleasure-in-agency similarly locates a reward in contributing rather than in passively receiving a public good. These approaches distinguish participation-contingent motivation from the policy benefit shared by participants and nonparticipants. The distinction supplies the strategic complementarity in the benchmark and clarifies the conditional role of the citizens’ policy prize.

Little (2016) distinguishes two coordination problems that potential protesters face. The first is “political coordination,” deciding whether to participate without knowing how many others are

dissatisfied enough to join. The second is “tactical coordination,” deciding when, where, and how to protest. His political coordination model also has a continuum of citizens with heterogeneous assessments and equilibria in cutoff strategies. He shows that better tactical information “almost always lowers the effective cost of protest, leading to larger (and more cohesive) antiregime action.” This distinction matters for the present model because communication technology informs citizens about the critical threshold they face here. I return to the point in the discussion of measurement and research design.

3 Modeling protests, repression, and policy change.

In this section, I develop a model of citizens who decide to protest to change a disliked policy. The citizens face a coordination and collective action problem. The protest changes the policy only if citizens’ participation reaches a given threshold. The threshold is an uncertain fundamental, and citizens receive noisy private signals of it. On the other side, the state would prefer not to change the policy. To influence participation, it can set the expected repression cost borne by a participant. A realized mass protest consumes operational resources, while a policy that imposes greater expected harm invites legal, electoral, and reputational backlash. I begin with the citizens’ participation problem and its complete-information benchmark. Next, I solve the global game. Then, I propose the two-stage sequential game in which the state sets expected participant cost first, and I inspect the strategic implications of the model.

3.1 The citizens’ participation problem.

There is a continuum of citizens of mass one, indexed by $i \in [0, 1]$. Each citizen chooses between participating in the protest (P) or not (NP). Let $n_i \in \{0, 1\}$ indicate participation, and let $N^* \in [0, 1]$ denote the realized mass of participants. Protesters want to change a policy so gridlocked that ordinary representative channels struggle to move it. The protest is successful if participation reaches a critical threshold τ , so the policy changes if and only if $N^* \geq \tau$.

The payoff primitives are as follows. A change in the policy provides a collective and indivisible benefit B_p to all individuals in the society (i.e., a public good). Individuals who participate may get an expressive or emotional benefit d_p . They may feel, for example, that fighting for a

Table 2: Notation and role in the model

Object	Meaning and units	Role
τ	Critical participating share required for policy change	Realized fundamental
N^*	Share of citizens who participate	Equilibrium outcome
B_p	Public policy benefit from successful protest	Citizen payoff primitive
d_p	Success-contingent participation reward	Citizen payoff primitive
q_r	Probability or share of participants targeted	Measured component of the directive
s_r	Harm conditional on being targeted	Measured component of the directive
$c_r = q_r s_r$	Expected repression cost per participant	State choice
B_s	Government value of preserving the policy status quo	State payoff primitive
$\tilde{N}(c_r)$	Expected share of citizens who participate	Anticipated outcome
$H(\tilde{N}, c_r; \theta_0)$	Planned operational cost for expected protest volume and the chosen repression policy	State cost technology
$K(c_r; \theta_1)$	Political cost of expected participant harm	State cost technology
σ	Noise in citizens' private signals	Information parameter

just cause is good in and of itself. Participants also pay c_r , the expected repression cost borne by a participant. This object combines the probability of victimization with the conditional harm suffered if victimized. If q_r denotes reach and s_r conditional severity, then $c_r = q_r s_r$.

Assumption 1 (One-dimensional directive). The executive chooses from an institution-specific, one-dimensional menu of policing directives indexed by c_r . Along this policy menu, reach is $q_r(c_r)$, conditional severity is $s_r(c_r)$, and $q_r(c_r)s_r(c_r) = c_r$.

The directive is the policy unit; q_r and s_r are its measurable components. This restriction fits institutions in which rules of engagement bundle deployment, targeting, and authorized force into stable policy packages. The functions H and K below evaluate costs along that feasible policy menu. Expected participant harm is therefore sufficient for the citizen's deterrence decision and indexes the packages compared by the executive. Conditional severity refers only to s_r .

Two features of the continuum setting must be made explicit, because they discipline how the primitives enter the participation margin.

First, each individual citizen is nonpivotal. With finitely many players, Δp_i would place the public good benefit B_p in the individual calculus. That term is the change in the probability of

meeting the threshold induced by one's own participation. In the continuum, a single citizen has measure zero, so $\Delta p_i = 0$. The citizen receives B_p when the protest succeeds whether she participated or abstained, and B_p cancels from that comparison. The benchmark therefore isolates the participation-contingent reward as the motive of the marginal participant.

Second, the expressive benefit must be success-contingent for participation to respond to the prospects of success at all. I follow the psychological rewards setting of Bueno de Mesquita and Shadmehr (2023), which builds on Wood's (2003) pleasure-in-agency. A participant enjoys the expressive reward d_p only if the movement succeeds. That satisfaction arrives only when the change actually happens. This success-contingency is a substantive assumption, and it generates the coordination problem. Were d_p unconditional, participation would reduce to the comparison of d_p against c_r , independent of any belief about others, and the strategic structure would disappear.

With these two features, the net payoff of participating relative to abstaining, for citizen i , is

$$u_i(P) - u_i(NP) = \mathbb{1}\{N^* \geq \tau\} \cdot d_p - c_r,$$

since $B_p \cdot \mathbb{1}\{N^* \geq \tau\}$ enters both sides and cancels. A citizen therefore participates when her belief that the protest will succeed, times the expressive reward, exceeds the expected repression cost.

3.2 The complete-information benchmark.

Suppose first, as a benchmark, that τ is common knowledge. Assume $0 < \tau \leq 1$, the interesting range, and $c_r < d_p$, so that participation is worthwhile when success is assured. Then two pure strategy equilibria coexist, and the small-noise limit selects a unique interior prediction (Carlsson and van Damme 1993; Morris and Shin 1998, 2003). If every citizen participates, $N^* = 1 \geq \tau$, the protest succeeds, and each participant receives $d_p - c_r > 0$. No citizen gains by deviating, since abstaining forfeits $d_p - c_r$ while leaving success (and hence B_p) unchanged. If no citizen participates, $N^* = 0 < \tau$, the protest fails, and a lone deviator would pay c_r for a failed protest. Universal abstention is therefore also an equilibrium. If instead $c_r > d_p$, universal abstention is the unique equilibrium, because participation is dominated even under assured success.

This benchmark mirrors the complete-information benchmark of Bueno de Mesquita and Shadmehr (2023, their Proposition 1), and it inherits the two problems they identify. The multi-

plicity makes prediction difficult, and the equilibrium set is insensitive to the parameters, including the repression policy itself. One could respond by selecting an equilibrium, for instance the mixed strategy equilibrium, and reading its participation probabilities as frequencies. I instead take seriously that the critical threshold is a variable citizens cannot observe. The size a protest must reach to force a policy change is written nowhere, unlike the threshold to win an election under plurality rule, which is common knowledge. How citizens form beliefs about τ is therefore central to any behavioral prediction.

3.3 The global game.

Let the critical threshold τ be drawn from the common prior $U[0,1]$ and remain unobserved by citizens. Citizens and the state use this same proper prior. Each citizen receives a noisy private signal

$$x_i = \tau + \sigma \varepsilon_i,$$

where the ε_i are i.i.d. with a log-concave density f and c.d.f. F with full support on $\mathbb{R}$, independent of τ , and $\sigma > 0$ scales the noise. A citizen with a low signal believes the required crowd is small and success likely. A citizen with a high signal believes success is out of reach.

A strategy is a mapping from signals to participation. The participation payoff is decreasing in the (believed) threshold, so cutoff strategies are the natural place to look for equilibria. A citizen participates if and only if $x_i < \hat{x}$. Given such a cutoff, the mass of participants at true threshold τ is

$$N^*(\tau) = \Pr(x_i < \hat{x} \mid \tau) = F\left(\frac{\hat{x} - \tau}{\sigma}\right),$$

which is continuous and strictly decreasing in τ . The protest therefore succeeds if and only if τ is below a unique critical value $\hat{\tau}$. The equilibrium is characterized by the same two conditions as in the canonical regime change game (Bueno de Mesquita and Shadmehr 2023, their equation (1)):

$$\text{(belief consistency / critical mass):} \quad F\left(\frac{\hat{x} - \hat{\tau}}{\sigma}\right) = \hat{\tau},$$

$$\text{(indifference of the marginal citizen):} \quad \Pr(\tau \leq \hat{\tau} \mid x_i = \hat{x}) \cdot d_p = c_r.$$

The first condition says that at the critical threshold the crowd is exactly critical. The second says

that a citizen whose signal equals the cutoff is indifferent between participating and abstaining. Under the proper prior, the marginal citizen's exact posterior success probability is

$$\pi_\sigma(\hat{x}, \hat{\tau}) = \frac{F(\hat{x}/\sigma) - F((\hat{x} - \hat{\tau})/\sigma)}{F(\hat{x}/\sigma) - F((\hat{x} - 1)/\sigma)}.$$

Thus a positive-noise monotone cutoff equilibrium solves

$$F\left(\frac{\hat{x}_\sigma - \hat{\tau}_\sigma}{\sigma}\right) = \hat{\tau}_\sigma, \quad d_p \pi_\sigma(\hat{x}_\sigma, \hat{\tau}_\sigma) = c_r. \quad (1)$$

These equations retain the boundary terms generated by $U[0,1]$. At an interior threshold they converge, as $\sigma \rightarrow 0$, to $(1 - \hat{\tau})d_p = c_r$. The limiting success threshold is therefore

$$\hat{\tau}(c_r) = \max\left\{1 - \frac{c_r}{d_p}, 0\right\}. \quad (2)$$

This is the analog of the limiting threshold $\theta^p = 1 - c$ in Bueno de Mesquita and Shadmehr (2023, their equation (2) and Proposition 2), with normalized expected participant harm c_r/d_p in the role of c . Proposition 1 states the proper-prior convergence result.

Proposition 1 (Proper-prior small-noise selection). *Fix a policy with $c_r/d_p \in (0,1)$. Consider any sequence of interior monotone cutoff equilibria $(\hat{x}_\sigma, \hat{\tau}_\sigma)$ satisfying equation (1), with $\hat{\tau}_\sigma$ bounded away from zero and one as $\sigma \rightarrow 0$. Every such sequence converges to*

$$\hat{x}_\sigma \longrightarrow \hat{\tau}, \quad \hat{\tau}_\sigma \longrightarrow \hat{\tau} = 1 - \frac{c_r}{d_p}.$$

The interior monotone limiting selection is therefore unique. If $c_r \geq d_p$, no citizen participates and the economically relevant limiting threshold is zero; if $c_r = 0$, every citizen participates and the limiting threshold is one.

The full proof is in the appendix. Three remarks.

First, every interior monotone equilibrium sequence converges to the same prediction for each policy. In the limiting game, protest succeeds if and only if $\tau \leq \hat{\tau}(c_r)$, so the state's comparative statics use the unique small-noise selection.

Second, the limiting outcome changes smoothly with the repression policy. The deterrence

derivative is¹

$$\frac{\partial \hat{\tau}}{\partial c_r} = -\frac{1}{d_p}.$$

Each unit of expected participant cost shrinks the set of fundamentals at which the protest succeeds at a constant rate, inversely proportional to the expressive reward. Repression is blunt against a strongly motivated citizenry (high d_p), and sharp against a weakly motivated one. This is the global games rendering of a familiar intuition: intrinsic motivation erodes the deterrent effect of expected participant harm.

Third, the direct deterrence margin is governed by d_p alone. The zero partial effect of B_p is a benchmark result conditional on three assumptions: citizens form a continuum, the policy prize is public and available equally to participants and abstainers, and policy stakes are separable from success-contingent motivation. A larger policy stake may still raise d_p by intensifying grievance, identity, efficacy, or pleasure-in-agency, and it may raise the government's B_s . The model isolates the direct pivotality channel by fixing those pathways.

The state's problem is evaluated in the small-noise limit $\sigma \rightarrow 0$. The cutoff $\hat{x}$ then converges to $\hat{\tau}$ and participation becomes sharp: $N^*(\tau) \rightarrow 1$ for $\tau < \hat{\tau}$ and $N^*(\tau) \rightarrow 0$ for $\tau > \hat{\tau}$. Expected protest mass is therefore $\bar{N}(c_r) \equiv \mathbb{E}_\tau[N^*(\tau) \mid c_r] = \hat{\tau}(c_r)$. The limiting environment captures the extensive margin between a mass protest and no mobilization and keeps the state's planning problem transparent.

3.4 The state's problem.

The elected executive knows the policy dispute and its stakes, then fixes an operational directive before citizens receive their private information and mobilize. Police implement that directive during the episode.

Assumption 2 (Within-episode commitment). The state chooses c_r after the policy dispute is known and before citizen mobilization, and c_r remains fixed during the protest episode.

The timing captures advance deployment, rules of engagement, authorized tactics, and expectations formed from directives and prior practice. The directive is publicly announced, legally

¹The notation $\partial \hat{\tau} / \partial c_r$ denotes the change in the limiting equilibrium success threshold induced by a marginal change in expected participant cost, with d_p fixed. The expression applies on the interior region $0 < c_r < d_p$; at the clamped boundary the relevant one-sided derivative is zero.

recorded, or credible under standing practice before citizens mobilize. Legal review, administrative compliance, press scrutiny, and anticipatory electoral response can therefore impose a political cost when the directive is authorized, even if deterrence prevents its use.

Let $H(\bar{N}, c_r; \theta_0)$ denote the operational cost of preparing for expected protest mass $\bar{N}$ under policy c_r . It includes personnel, equipment, transport, command capacity, and other resources committed in anticipation of protest, including preparedness expenditures incurred before turnout is realized.

Assumption 3 (Planned operational cost). The function $H(\bar{N}, c_r; \theta_0)$ is twice continuously differentiable on $\bar{N} \in [0, 1]$ and $c_r \geq 0$, satisfies $H(0, 0; \theta_0) = 0$, $H_N > 0$, $H_c \geq 0$, and $H_{N\theta_0} > 0$.

The shifter θ_0 raises the marginal cost of preparing for additional protesters. The cross-partial $H_{c\theta_0}$ permits the same shifter to change the direct operational cost of harsher tactics.²

The same subscript convention applies to the political-cost function.³

Assumption 4 (Political cost of the authorized directive). The function $K(c_r; \theta_1)$ is twice continuously differentiable on $c_r \geq 0$, satisfies $K(0; \theta_1) = 0$, $K_c(0; \theta_1) = 0$, $K_c(c_r; \theta_1) > 0$ for $c_r > 0$, $K_{cc}(c_r; \theta_1) > 0$, $K_{c\theta_1}(c_r; \theta_1) > 0$, and $K_c(c_r; \theta_1) \rightarrow \infty$ as $c_r \rightarrow \infty$.

Strict convexity captures rising marginal legal, electoral, administrative, and reputational costs of authorizing greater expected harm per participant. The cross-partial gives θ_1 a transparent institutional interpretation among democracies: a larger θ_1 raises the marginal political price of every positive c_r .

The state keeps $B_s > 0$ when the status quo survives; its policy payoff after protest-induced change is normalized to zero. With the proper planning distribution $\tau \sim U[0, 1]$, the success probability and expected protest mass in the small-noise limit both equal $\hat{\tau}(c_r)$. Expected utility is

$$u_S(c_r) = (1 - \hat{\tau}(c_r)) B_s - H(\hat{\tau}(c_r), c_r; \theta_0) - K(c_r; \theta_1). \quad (3)$$

²Subscripts on H denote partial derivatives evaluated at $(\bar{N}, c_r; \theta_0)$. Thus $H_N \equiv \partial H / \partial \bar{N}$, $H_c \equiv \partial H / \partial c_r$, $H_{NN} \equiv \partial^2 H / \partial \bar{N}^2$, $H_{Nc} \equiv \partial^2 H / (\partial \bar{N} \partial c_r)$, $H_{cc} \equiv \partial^2 H / \partial c_r^2$, $H_{N\theta_0} \equiv \partial^2 H / (\partial \bar{N} \partial \theta_0)$, and $H_{c\theta_0} \equiv \partial^2 H / (\partial c_r \partial \theta_0)$. Every partial derivative fixes the function's remaining arguments.

³Thus $K_c \equiv \partial K(c_r; \theta_1) / \partial c_r$, $K_{cc} \equiv \partial^2 K(c_r; \theta_1) / \partial c_r^2$, and $K_{c\theta_1} \equiv \partial^2 K(c_r; \theta_1) / (\partial c_r \partial \theta_1)$. The function's other argument remains fixed in each derivative.

For $c_r \in (0, d_p)$, the marginal payoff is⁴

$$u_c(c_r) = \frac{B_s + H_N}{d_p} - H_c - K_c. \quad (4)$$

The term $(B_s + H_N)/d_p$ values the status quo and planned operational resources saved when deterrence reduces expected protest mass. The term H_c is the direct operational cost of implementing harsher tactics. The term K_c is the marginal political cost of expected participant harm.

Assumption 5 (Interior curvature). The objective is twice continuously differentiable and strictly concave on $(0, d_p)$, with $u_{cc}(c_r) < 0$ and right derivative $u_c(0^+) > 0$.

Using equation (4), strict concavity is equivalent to

$$\frac{H_{NN}}{d_p^2} - \frac{2H_{Nc}}{d_p} + H_{cc} + K_{cc} > 0.$$

This condition allows planned operational cost to be nonlinear in expected protest mass and expected participant harm while keeping a unique policy response.

Proposition 2 (The optimal repression policy). *Under Assumptions 2–5, the state has a unique optimal policy on $[0, d_p]$. The solution is interior and satisfies $u_c(c_r^*) = 0$ when the left derivative $u_c(d_p^-) < 0$; otherwise the state chooses the full-deterrence corner $c_r^* = d_p$. Expected harm above d_p is dominated because the success threshold and expected protest mass are already clamped at zero while H and K are weakly increasing in c_r .*

3.5 Comparative statics.

The general technology separates effects that follow directly from the model from the condition governing the crowd-cost mechanism.

⁴Here $u_c \equiv \partial u_S / \partial c_r$ and $u_{cc} \equiv \partial^2 u_S / \partial c_r^2$. These derivatives vary the state's policy c_r and allow the induced expected mass $\tilde{N}(c_r) = \hat{\tau}(c_r)$ to adjust, while fixing B_s , d_p , θ_0 , θ_1 , and the functional forms of H and K .

Proposition 3 (Comparative statics). *At the interior solution,*⁵

$$\begin{aligned}\frac{\partial c_r^*}{\partial B_s} &= -\frac{1/d_p}{u_{cc}} > 0, & \frac{\partial c_r^*}{\partial \theta_1} &= \frac{K_{c\theta_1}}{u_{cc}} < 0, \\ \frac{\partial c_r^*}{\partial d_p} &= \frac{B_s + H_N + c_r^* H_{Nc} - (c_r^*/d_p) H_{NN}}{d_p^2 u_{cc}}, & \frac{\partial c_r^*}{\partial B_p} &= 0,\end{aligned}$$

Thus $\partial c_r^*/\partial d_p < 0$ when $B_s + H_N + c_r^* H_{Nc} > (c_r^*/d_p) H_{NN}$. For the crowd-cost shifter,

$$\frac{\partial c_r^*}{\partial \theta_0} = -\frac{H_{N\theta_0}/d_p - H_{c\theta_0}}{u_{cc}}. \quad (5)$$

Hence a higher marginal cost of preparing for protesters increases expected participant harm if and only if

$$\frac{H_{N\theta_0}}{d_p} > H_{c\theta_0}. \quad (\text{Crowd-cost condition})$$

A rise in B_s lowers the equilibrium success probability, whereas a rise in θ_1 raises it. When the crowd-cost condition holds, a rise in θ_0 lowers the success probability. A rise in d_p raises it whenever the stated condition for $\partial c_r^*/\partial d_p < 0$ holds.

The crowd-cost condition is the paper's central technological result. The term $H_{N\theta_0}/d_p$ is the additional planning cost saved when deterrence reduces expected protest mass. The term $H_{c\theta_0}$ captures how the same shifter changes the direct operational cost of harsher tactics. The first force raises the state's optimal expected harm per participant; the second tempers that increase and can reverse it.

Two organizational technologies give the condition substantive content. Let w denote the price of personnel and logistical inputs, and let $A(c_r)$ collect operational costs that do not vary with w . In a *deployment-separable* system,

$$H^D(\bar{N}, c_r; w) = w\bar{N} + A(c_r). \quad (6)$$

⁵Each expression $\partial c_r^*/\partial z$ or $\partial \hat{\tau}^*/\partial z$ is an equilibrium comparative-static derivative. It allows the optimizing policy, expected protest mass, and success threshold to adjust to primitive z while the other listed primitives and both cost-function families remain fixed. All derivatives of H on the right-hand side are evaluated at $(\bar{N}, c_r; \theta_0) = (\hat{\tau}^*, c_r^*; \theta_0)$; derivatives of K are evaluated at $(c_r^*; \theta_1)$.

The input price raises the marginal cost of preparing for an additional protester, $H_{Nw}^D = 1$, without changing the direct marginal cost of the directive, $H_{cw}^D = 0$. Thus $\partial c_r^*/\partial w > 0$: higher personnel, transport, or crowd-control deployment costs make deterrence more valuable.⁶

In a *tactic-intensive* system, the same inputs also implement the authorized level of expected harm:

$$H^T(\bar{N}, c_r; w) = w(\bar{N} + \lambda c_r) + A(c_r), \quad \lambda > 0. \quad (7)$$

Now $H_{Nw}^T = 1$ and $H_{cw}^T = \lambda$. The price increase raises expected participant harm when $\lambda < 1/d_p$ and lowers it when $\lambda > 1/d_p$. The second case describes organizations in which specialized officers, less-lethal munitions, medical support, detention processing, or documentation requirements make stronger directives especially input-intensive. Equations (6) and (7) yield a discriminating implication: comparable input-price increases lead the state to authorize more expected participant harm where they load mainly on personnel deployment and less expected participant harm where they load heavily on implementing force.

The institutional comparison follows directly from $K_{c\theta_1} > 0$. Environments with stronger legal liability, judicial oversight, electoral accountability, or media scrutiny raise the marginal political price of expected harm and therefore reduce c_r^* . This gives the democratic setting a formal role through the cost schedule faced by the elected executive.

For the transparent benchmark, use

$$H(\bar{N}, c_r; \theta_0) = \theta_0 \bar{N}, \quad K(c_r; \theta_1) = \theta_1 c_r^2.$$

This is the deployment-separable technology with $A(c_r) = 0$. Then $H_{N\theta_0} = 1$, $H_{c\theta_0} = 0$, the crowd-cost condition holds strictly, and the solution reduces to

$$c_r^* = \min \left\{ \frac{B_s + \theta_0}{2\theta_1 d_p}, d_p \right\}, \quad \hat{\tau}^* = \max \left\{ 1 - \frac{B_s + \theta_0}{2\theta_1 d_p^2}, 0 \right\}.$$

Figure 1 uses this additive-quadratic specialization to show the mechanism.

In the benchmark, a higher θ_0 raises expected participant harm. It increases the planned

⁶The derivative $\partial c_r^*/\partial w$ allows the optimal directive and anticipated protest mass to adjust to the input price w ; B_s , d_p , θ_1 , and the functional forms of A and K remain fixed. The other arguments of H remain fixed in the cross-partial H_{Nw} and H_{cw} .

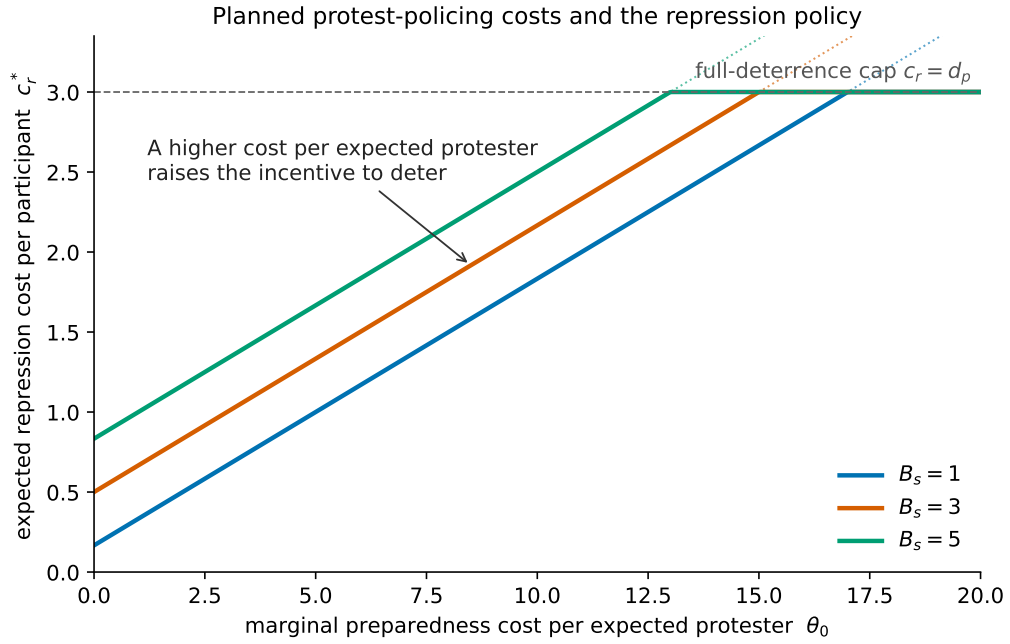


Figure 1: The crowd-management mechanism in the additive-quadratic benchmark, $H(\bar{N}, c_r; \theta_0) = \theta_0 \bar{N}$ and $K(c_r; \theta_1) = \theta_1 c_r^2$, with $\theta_1 = 1$ and $d_p = 3$. Solid lines show the effective policy for three values of the government's status quo stake; dotted lines show the unconstrained interior solution. A higher marginal cost of preparing for protesters raises expected participant harm until the full-deterrence cap binds.

resources saved by reducing expected protest mass without changing the direct operational cost of harsher tactics. A higher θ_1 lowers c_r by steepening the marginal political-cost schedule. Greater intrinsic motivation d_p lowers the interior policy because motivated citizens are harder to deter. At the full-deterrence corner, c_r tracks d_p one for one.

Two features of these results deserve emphasis.

First, the status quo prize B_s has an unambiguously positive effect at every interior optimum. The monotone cutoff equilibrium makes the deterrence technology the linear relation (2), so the marginal value of expected participant harm rises with the government's stake in the disputed policy.

Second, the direct partial effect of B_p equals zero under the benchmark's nonpivotality and separability assumptions. Policy stakes affect repression through the participation reward d_p and the government's status quo value B_s . The relevant distinction is between the prize's direct public-good component and the motivational and governmental channels through which politically important disputes alter behavior.

3.6 Directive costs and exposure-dependent political costs.

The baseline $K(c_r; \theta_1)$ is paid when the executive authorizes an observable directive. Democratic institutions may also impose consequences after citizens are exposed to it. In the small-noise limit, total expected harm in the population is

$$T(c_r) = c_r \hat{t}(c_r).$$

Let $L(T; \theta_2)$ be an increasing, convex political cost of exposure, in addition to the directive cost K . The state then maximizes

$$u_S^E(c_r) = (1 - \hat{t})B_s - H(\hat{t}, c_r; \theta_0) - K(c_r; \theta_1) - L(c_r \hat{t}; \theta_2).$$

At an interior policy its marginal payoff is⁷

$$u_c^E(c_r) = \frac{B_s + H_N}{d_p} - H_c - K_c - L_T \left(\hat{t} - \frac{c_r}{d_p} \right). \quad (8)$$

Exposure-dependent accountability changes the optimal level because total expected harm first rises and then falls as deterrence reduces the crowd. Differentiating equation (8) in the planning-cost shifter again gives $H_{N\theta_0}/d_p - H_{c\theta_0}$ when the extended objective remains strictly concave and L does not depend on θ_0 . The crowd-cost condition therefore continues to determine the sign of $\partial c_r^*/\partial \theta_0$. The directive cost remains positive at full deterrence, while the exposure cost reaches zero with expected turnout.

3.7 An extension: Repression, protests, and emotions.

The benchmark holds success-contingent motivation fixed. To isolate one feedback, let citizens observe the directive c_r before choosing whether to participate and let $d_p(c_r) = d_0 + e(c_r)$. The observable object can be a public rule, announced tactic, or credible expectation formed from prior police practice. The threshold becomes

$$\hat{t}(c_r) = \max \left\{ 1 - \frac{c_r}{d_0 + e(c_r)}, 0 \right\}.$$

At an interior threshold, differentiation gives⁸

$$\frac{\partial \hat{t}}{\partial c_r} = - \frac{d_0 + e(c_r) - c_r e'(c_r)}{(d_0 + e(c_r))^2}.$$

Define the elasticity of success-contingent motivation to expected participant harm as $\varepsilon(c_r) = c_r e'(c_r)/d_p(c_r)$.

Proposition 4 (Deterrence and backlash). *Expected participant harm deters, $\partial \hat{t}/\partial c_r < 0$, if and only if $\varepsilon(c_r) < 1$; it backfires, $\partial \hat{t}/\partial c_r > 0$, if and only if $\varepsilon(c_r) > 1$.*

The extension converts the punishment puzzle into a scale-free condition. Anticipated repres-

⁷Here $L_T \equiv \partial L(T; \theta_2)/\partial T$, evaluated at $T = c_r \hat{t}(c_r)$. The derivative varies c_r , allows $\hat{t}(c_r)$ and total expected harm to adjust, and fixes $B_s, d_p, \theta_0, \theta_1, \theta_2$, and the cost-function families. Since $\partial \hat{t}/\partial c_r = -1/d_p$, $\partial(c_r \hat{t})/\partial c_r = \hat{t} - c_r/d_p$.

⁸The derivative $\partial \hat{t}/\partial c_r$ varies the observable policy c_r and fixes d_0 and the response function $e(\cdot)$; $e'(c_r)$ is the ordinary derivative of that function evaluated at c_r .

sion deters when motivation rises less than proportionally with expected harm and recruits when it rises more than proportionally. Evidence that mobilization follows a coercive tactic is informative about this mechanism when the tactic causally changes both expected participant harm and success-contingent motivation. The timing concerns an observable ex ante policy. Across repeated episodes, prior realized violence can determine citizens' current expectations of that policy.

4 Discussion: contribution, observables, and research design.

The theory distinguishes the planned operational cost of expected protest volume from the political cost of authorizing expected harm per participant. These costs affect the state's choice differently. A higher marginal cost of preparing for protesters makes deterrence more valuable, whereas the marginal legal, electoral, administrative, and reputational costs of the directive restrain the policy used to deter. The derived crowd-cost condition determines the direction of the crowd-cost effect when the same cost environment also changes the direct operational cost of harsher tactics. Expected participant harm rises only when the planning resources saved through deterrence exceed that additional operational cost. Equation (8) adds political consequences from total expected harm while preserving this sign condition.

The model assigns separate roles to three political quantities. The government's B_s is the value of preserving the disputed status quo. Citizens' d_p is the participation-contingent reward from contributing to a successful movement. The public policy benefit B_p has no direct pivotality effect in the continuum benchmark, but politically important disputes can affect both d_p and B_s . Klein and Regan's (2018) cross-national protest-event analysis provides suggestive evidence about status quo stakes. They find that high concession costs, meaning demands that are expensive for the ruling coalition to accept, are associated with coercive state responses. This pattern is consistent with the positive effect of B_s . They also find that high disruption costs encourage accommodation. That result points in the opposite direction from the additive crowd-cost benchmark, but it concerns economic and public-order losses tied to a protest's location, duration, and size. The model's θ_0 instead changes the marginal operational cost of preparing for an additional participant. The contrast shows why the two costs should not be combined into a single measure of how expensive a protest is for the government.

The observable counterpart of c_r combines reach and intensity. Let q_r be the probability that a participant is targeted and s_r the harm conditional on targeting. An event-level measure of expected participant harm joins participant exposure to evidence about deaths, injuries, detention, sexual violence, psychological trauma, and other coercive tactics, so that $c_r = q_r s_r$. Under Assumption 1, observed directives trace an institution-specific policy menu. The executive does not choose q_r and s_r independently. Measuring both components constructs the deterrence index and reveals how police organizations package reach and conditional severity. Castro (2025) shows the value of tactic-specific information by coding civilian complaints during Chile’s 2019 protest cycle by tactic, municipality, and day and validating them against media reports and police ammunition records. She finds that selective high-severity tactics generated short-run mobilization and that low-severity tactics increased protest regardless of selectivity. Those results are consistent with the backfire region of Proposition 4 for the observed tactic combinations, although they do not estimate the model’s motivational elasticity.

Estimating the operational cost $H(\bar{N}, c_r; \theta_0)$ requires the government’s anticipated protest volume and the resources committed before an episode. The relevant inputs include scheduled personnel, overtime capacity, transport, equipment and consumables, temporary command infrastructure, detention capacity, and medical support. Benchmarking their prices against comparable labor and procurement costs measures opportunity cost across jurisdictions. Budget documents, deployment plans, procurement records, and after-action accounts can distinguish standing capacity from resources added in anticipation of protest. They can also estimate the loading λ in equation (7): input-price changes that mainly raise deployment cost predict greater expected participant harm, while price changes that strongly raise the cost of implementing force predict restraint. Within-jurisdiction changes are especially informative because ordinary crime also affects resource demand and police capacity.

Institutions shift $K(c_r; \theta_1)$ by changing the consequences of authorizing a directive for elected officials. Published rules of engagement, judicial preclearance, legislative oversight, internal compliance review, and press access expose the authorization itself to scrutiny. Civil liability, casualty investigations, and electoral reactions to documented injuries instead load on $L(T; \theta_2)$. Aytaç, Schiumerini, and Stokes (2017) study backlash movements in Turkey, Brazil, and Ukraine and find that governments with stable electoral support were more willing to use harsh measures, whereas

electorally insecure governments had stronger incentives to reduce violence. This pattern is consistent with the model's claim that a lower marginal electoral cost permits greater expected participant harm. At a broader level, Davenport and Armstrong (2004) find that democracy reduces human-rights violations only above a threshold of democratic development. Their country-level outcome is not specific to protest directives, but the nonlinear result is consistent with interpreting θ_1 as meaningful institutional variation among democracies. Research designs can use the timing of these institutions to distinguish changes in the marginal directive cost K_c from changes in the marginal exposure cost L_T .

Existing evidence also points to the motivational channel, while showing that participants need not respond uniformly. Adam-Troian, Çelebi, and Mahfud's (2020) cross-sectional study of 523 Yellow Vest protesters finds that exposure to police violence is positively associated with intentions to attend future demonstrations and to sacrifice for the movement. Perceived loss of significance and identification with the Yellow Vests mediate those associations. Sullivan and Davenport's (2017) panel study of the Republic of New Africa finds organizational backlash but different individual responses: members who personally experienced repression became more likely to participate, while members who were not directly repressed withdrew. These findings are consistent with a positive motivational response for some citizens, but they also caution against treating $e(c_r)$ as homogeneous. Survey and experimental data can measure d_p and the elasticity in Proposition 4 more directly. Items can measure political efficacy, anticipated satisfaction from contributing to collective success, and willingness to participate under stated probabilities of success and victimization. A factorial design can vary victimization reach, conditional harm, and the probability of policy success separately. The resulting responses distinguish the direct deterrent effect of expected harm from any change in success-contingent motivation.

Event records can first describe the joint variation in anticipated turnout, planned operational inputs, coercive reach, conditional harm, and institutional constraints. Suitable designs can then use institutional reforms and predetermined changes in labor or procurement costs to isolate particular margins. The comparative statics specify the expected pattern. Status quo stakes raise expected participant harm, and stronger marginal accountability lowers it. A planning-cost shifter raises expected harm only where the resources saved through deterrence exceed the direct operational-cost effect identified by the crowd-cost condition.

5 Conclusion.

This paper explains why a government that finds large protests expensive to manage may threaten harsher repression before people mobilize. The logic has two steps. Expected harm discourages some citizens from joining. A smaller expected crowd lets the government plan fewer officers, vehicles, detention spaces, and supplies. Harsher repression becomes attractive when the resources saved by preparing for a smaller crowd exceed the added operational and political costs of authorizing force. What matters is the cost of managing one more protester. The total price of maintaining a police force is a different quantity.

This tradeoff also explains why governments facing similar protests can respond differently. Consider an economic downturn that cuts tax revenue while raising demand for unemployment benefits, emergency transfers, and other social spending. The resulting fiscal pressure leaves less money for police overtime, recruitment, and equipment. If officials still expect a large protest, they may be unable to expand deployment as they otherwise would. Where revising the use-of-force directive is comparatively cheap, the executive may compensate by authorizing tactics that expose each participant to greater expected harm, hoping that stronger deterrence allows a smaller force to manage the episode. The result still depends on how the police organization produces force. If stronger tactics require specialized officers, equipment, documentation, medical support, or detention capacity, the budget pressure can instead produce restraint. Legal review, political oversight, and electoral accountability further change the price that an elected government pays when it authorizes expected harm. Repression depends on which resources have become scarce and on how the directive uses them.

Citizens complete the argument because the crowd is not fixed in advance. Each person weighs the expected harm of participation against the satisfaction of helping a protest succeed. A stronger directive usually discourages participation and makes policy change less likely. Anger can reverse that result, but only when the added motivation grows faster than the expected harm. Repression can therefore anger citizens without producing a larger protest; anger and backfire are separate outcomes.

The argument points to a dynamic question for future research. Governments announce directives and plan deployments before an episode, but commanders and citizens also learn after

turnout and injuries become visible. Comparing the initial directive with later tactical revisions would show when governments use an early threat to avoid a costly crowd, when operational demands force restraint, and when observed violence creates the mobilization that the directive was meant to prevent.

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6 Appendix.

6.1 Full comparative-statics map for the additive-quadratic benchmark.

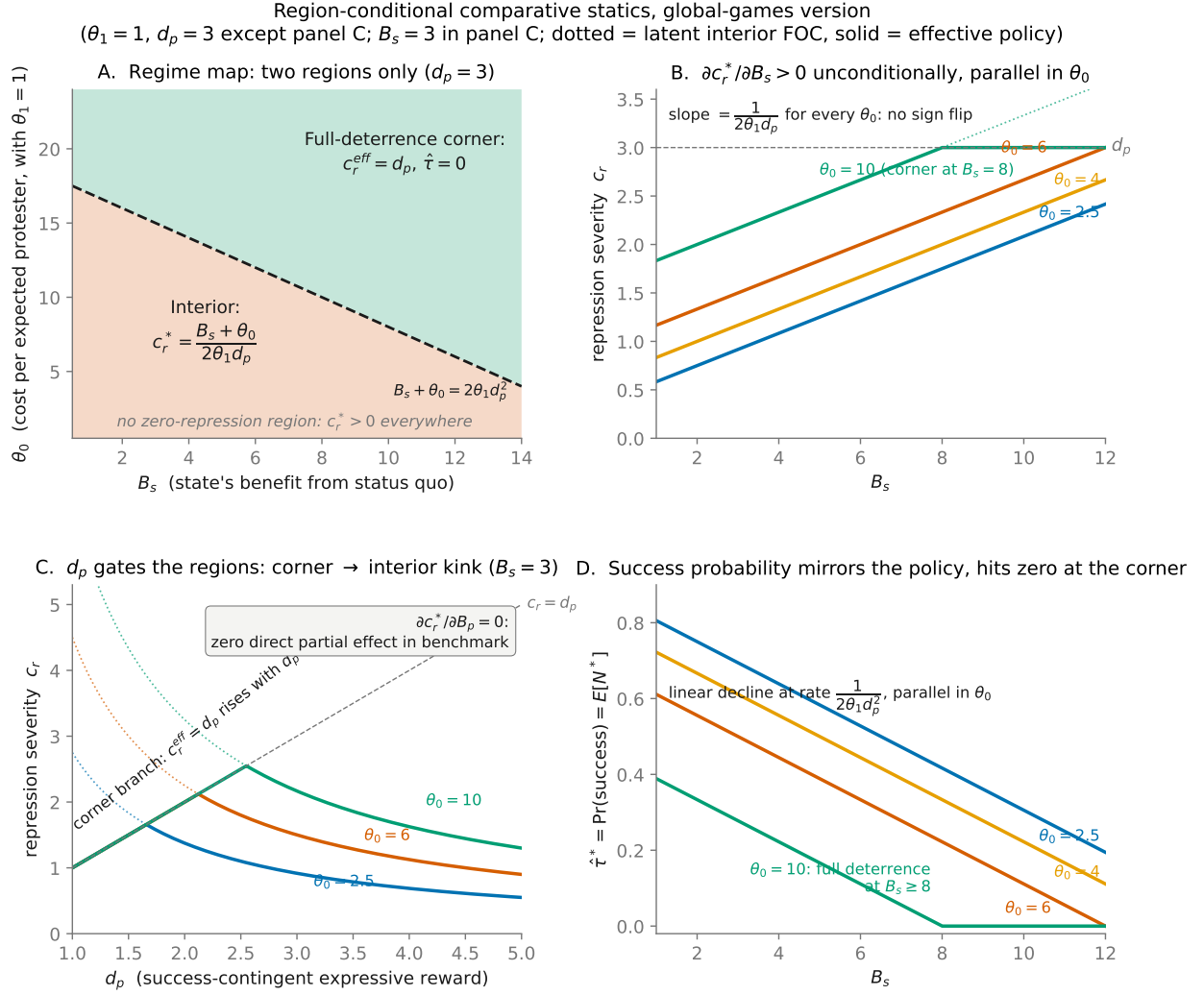


Figure 2: Region-conditional comparative statics for $H(\bar{N}, c_r; \theta_0) = \theta_0 \bar{N}$ and $K(c_r; \theta_1) = \theta_1 c_r^2$. Panel A maps the interior and full-deterrence regions. Panel B varies the government's status quo stake. Panel C varies success-contingent motivation. Panel D shows the corresponding probability of policy change. Dotted curves are unconstrained interior solutions and solid curves are effective policies.

6.2 Posterior beliefs under the common proper prior.

Lemma 1 (Proper-prior posterior). Suppose $\tau \sim U[0, 1]$ and $x_i = \tau + \sigma \varepsilon_i$, where ε_i has density f and c.d.f. F . For $t \in [0, 1]$,

$$\Pr(\tau \leq t \mid x_i = x) = \frac{F(x/\sigma) - F((x-t)/\sigma)}{F(x/\sigma) - F((x-1)/\sigma)}.$$

Proof. The likelihood of x conditional on τ is $f((x-\tau)/\sigma)/\sigma$. The uniform prior restricts its support to $[0, 1]$. Substituting $u = (x - \tau)/\sigma$ gives

$$\begin{aligned} \int_0^t \frac{1}{\sigma} f\left(\frac{x-\tau}{\sigma}\right) d\tau &= F(x/\sigma) - F((x-t)/\sigma), \\ \int_0^1 \frac{1}{\sigma} f\left(\frac{x-\tau}{\sigma}\right) d\tau &= F(x/\sigma) - F((x-1)/\sigma). \end{aligned}$$

Their ratio is the posterior probability. □

6.3 Proof of Proposition 1.

Proof of Proposition 1. Fix a cutoff strategy profile with cutoff $\hat{x}$: citizen i participates if and only if $x_i < \hat{x}$.

Step 1: participation and the induced success threshold. At true threshold τ , the mass of participants is

$$N^*(\tau) = \Pr(x_i < \hat{x} \mid \tau) = \Pr(\tau + \sigma \varepsilon_i < \hat{x}) = \Pr\left(\varepsilon_i < \frac{\hat{x} - \tau}{\sigma}\right) = F\left(\frac{\hat{x} - \tau}{\sigma}\right),$$

which is continuous and strictly decreasing in τ . At $\tau = 0$, $N^*(0) > 0$, while at $\tau = 1$, $N^*(1) < 1$. Hence $N^*(\tau) - \tau$ crosses zero exactly once on $(0, 1)$. Denote the crossing by $\hat{\tau}_\sigma$. It satisfies

$$F\left(\frac{\hat{x}_\sigma - \hat{\tau}_\sigma}{\sigma}\right) = \hat{\tau}_\sigma, \tag{9}$$

and protest succeeds exactly when $\tau \leq \hat{\tau}_\sigma$.

Step 2: the exact positive-noise system. A citizen with signal x_i gains $\Pr(\tau \leq \hat{\tau}_\sigma \mid x_i)d_p - c_r$ from participating. Lemma 1, evaluated at $t = \hat{\tau}_\sigma$ and $x_i = \hat{x}_\sigma$, gives

$$\Pr(\tau \leq \hat{\tau}_\sigma \mid x_i = \hat{x}_\sigma) = \frac{F(\hat{x}_\sigma/\sigma) - F((\hat{x}_\sigma - \hat{\tau}_\sigma)/\sigma)}{F(\hat{x}_\sigma/\sigma) - F((\hat{x}_\sigma - 1)/\sigma)}.$$

Together with marginal indifference, this expression and equation (9) give the exact system in equation (1).

Step 3: the interior small-noise limit. Define $z_\sigma = (\hat{x}_\sigma - \hat{\tau}_\sigma)/\sigma$. The crossing condition implies $F(z_\sigma) = \hat{\tau}_\sigma$ and therefore

$$\hat{x}_\sigma = \hat{\tau}_\sigma + \sigma F^{-1}(\hat{\tau}_\sigma).$$

Because $\hat{\tau}_\sigma$ remains in a compact subset of $(0, 1)$, $F^{-1}(\hat{\tau}_\sigma)$ is bounded. Hence $\hat{x}_\sigma - \hat{\tau}_\sigma \rightarrow 0$, $\hat{x}_\sigma/\sigma \rightarrow \infty$, and $(\hat{x}_\sigma - 1)/\sigma \rightarrow -\infty$. The marginal citizen's posterior success probability therefore converges to

$$\frac{1 - F(z_\sigma)}{1} = 1 - \hat{\tau}_\sigma.$$

Marginal indifference implies that every convergent subsequence satisfies $(1 - \hat{\tau})d_p = c_r$. Its only interior solution is $\hat{\tau} = 1 - c_r/d_p$. Every subsequence has the same limit, so the sequence converges, and $\hat{x}_\sigma \rightarrow \hat{\tau}$ as well.

Step 4: boundary policies. If $c_r \geq d_p$, the participation gain satisfies $\Pr(\cdot)d_p - c_r \leq 0$ for every belief, with strict inequality whenever the belief is interior, so no citizen participates. If $c_r = 0$, participation is weakly dominant and every citizen participates. These policies produce the limiting thresholds zero and one, respectively. $\square$

Remark 1. At positive noise, equation (1) is a two-dimensional system and its posterior contains both boundaries of the common prior. The boundaries vanish locally around an interior threshold as noise goes to zero. The uniqueness claim in Proposition 1 concerns this limiting selection: every interior monotone equilibrium sequence has the same limit. This is the selection used in the state's problem.⁹

6.4 Positive-noise numerical check.

The archived verification code solves equation (1) under Gaussian noise and follows the interior branch selected by Proposition 1. It also computes expected protest mass by numerical integration rather than imposing equality with the success probability. For $B_s = 3$, $d_p = 3$, $\theta_1 = 1$, and $H(\bar{N}, c_r; \theta_0) = \theta_0 \bar{N}$, the resulting policies are:

⁹The locally diffuse limiting argument follows the standard global-games logic in Morris and Shin (2003) and Bueno de Mesquita and Shadmehr (2023).

σ	c_r^* at $\theta_0 = 1.0$	c_r^* at $\theta_0 = 1.1$
0.05	0.6947	0.7134
0.03	0.6832	0.7013
0.02	0.6778	0.6954
0.01	0.6722	0.6894

The limiting policies are 0.6667 and 0.6833, respectively. The positive planning-cost effect is present at every reported noise level and the policies approach their analytic limits. The expected-mass approximation also tightens with noise: at $c_r/d_p = 0.2$, the absolute gap between expected mass and the success probability falls from 0.0421 at $\sigma = 0.05$ to 0.0084 at $\sigma = 0.01$.

6.5 Proof of Proposition 2.

Proof of Proposition 2. For $c_r \in [0, d_p]$, substitute $\hat{c}(c_r) = 1 - c_r/d_p$ into equation (3):

$$u_S(c_r) = \frac{c_r}{d_p} B_s - H\left(1 - \frac{c_r}{d_p}, c_r; \theta_0\right) - K(c_r; \theta_1).$$

Differentiation gives

$$\begin{aligned} u_c(c_r) &= \frac{B_s + H_N}{d_p} - H_c - K_c, \\ u_{cc}(c_r) &= -\frac{H_{NN}}{d_p^2} + \frac{2H_{Nc}}{d_p} - H_{cc} - K_{cc}. \end{aligned}$$

Assumption 5 makes the objective strictly concave on $(0, d_p)$ and gives $u_c(0^+) > 0$. If $u_c(d_p^-) < 0$, continuity of the interior derivative and strict concavity imply a unique root of $u_c(c_r) = 0$ in $(0, d_p)$, which is the unique maximizer. If $u_c(d_p^-) \geq 0$, the objective is increasing up to the full-deterrence corner and is maximized at $c_r = d_p$ on this interval. At every $c_r > d_p$, the clamped threshold and expected protest mass are zero, so $u_S(c_r) = B_s - H(0, c_r; \theta_0) - K(c_r; \theta_1)$ and its derivative is $-H_c(0, c_r; \theta_0) - K_c(c_r; \theta_1) < 0$. Thus the policy characterized in the proposition is globally optimal. This argument uses one-sided derivatives at the kink $c_r = d_p$. $\square$

6.6 Proof of Proposition 3.

Proof of Proposition 3. At an interior solution define¹⁰

$$G(c_r; B_s, \theta_0, \theta_1, d_p) \equiv u_c(c_r) = 0.$$

Because $G_{c_r} = u_{cc} < 0$, the implicit function theorem gives

$$\frac{\partial c_r^*}{\partial B_s} = -\frac{1/d_p}{u_{cc}} > 0, \quad \frac{\partial c_r^*}{\partial \theta_1} = \frac{K_{c\theta_1}}{u_{cc}} < 0,$$

where $K_{c\theta_1} > 0$. Fixing c_r and differentiating G in d_p gives

$$G_{d_p} = -\frac{B_s + H_N + c_r H_{Nc} - (c_r/d_p) H_{NN}}{d_p^2},$$

and therefore

$$\frac{\partial c_r^*}{\partial d_p} = \frac{B_s + H_N + c_r^* H_{Nc} - (c_r^*/d_p) H_{NN}}{d_p^2 u_{cc}}.$$

This derivative is negative under the condition stated in Proposition 3. For the crowd-cost shifter,

$$G_{\theta_0} = \frac{H_{N\theta_0}}{d_p} - H_{c\theta_0}, \quad \frac{\partial c_r^*}{\partial \theta_0} = -\frac{H_{N\theta_0}/d_p - H_{c\theta_0}}{u_{cc}}.$$

Since $u_{cc} < 0$, this derivative is positive exactly when the numerator is positive. This is the crowd-cost condition. The primitive B_p is absent from G , which gives its zero direct partial effect.

Finally, $\hat{\tau}^* = 1 - c_r^*/d_p$. Its response to B_s , θ_0 , and θ_1 has the opposite sign from the corresponding response of c_r^* . For motivation,

$$\frac{\partial \hat{\tau}^*}{\partial d_p} = \frac{c_r^*}{d_p^2} - \frac{1}{d_p} \frac{\partial c_r^*}{\partial d_p} > 0,$$

because $c_r^* > 0$ and $\partial c_r^*/\partial d_p < 0$. The direct partial effect of B_p on $\hat{\tau}^*$ is also zero. □

¹⁰For the first-order-condition function G , $G_z \equiv \partial G/\partial z$ fixes every displayed primitive other than z . In particular, $G_{c_r} = u_{cc}$, while G_{d_p} and G_{θ_0} differentiate the first-order condition at a fixed candidate policy c_r . In the limiting state problem, expected mass equals $\hat{\tau} = 1 - c_r/d_p$, so G_{d_p} includes the induced change $\partial \hat{\tau}/\partial d_p = c_r/d_p^2$ at fixed c_r . The implicit-function step then allows the equilibrium policy to adjust.

6.7 Proof of Proposition 4.

Proof of Proposition 4. With $d_p(c_r) = d_0 + e(c_r)$, apply the quotient rule to $\hat{\tau}(c_r) = 1 - c_r/d_p(c_r)$:

$$\frac{\partial \hat{\tau}}{\partial c_r} = -\frac{d_p(c_r) \cdot 1 - c_r e'(c_r)}{d_p(c_r)^2} = -\frac{d_p(c_r) - c_r e'(c_r)}{d_p(c_r)^2}.$$

The denominator is positive, so the derivative has the sign of $-(d_p(c_r) - c_r e'(c_r))$. Dividing the parenthesis by $d_p(c_r) > 0$,

$$d_p(c_r) - c_r e'(c_r) \geq 0 \iff 1 - \frac{c_r e'(c_r)}{d_p(c_r)} = 1 - \varepsilon(c_r) \geq 0,$$

so $\partial \hat{\tau} / \partial c_r < 0$ exactly when $\varepsilon(c_r) < 1$ and $\partial \hat{\tau} / \partial c_r > 0$ exactly when $\varepsilon(c_r) > 1$. □

The archived verification scripts also check the comparative statics for the planned-cost technology, the additive-quadratic specialization, and the expected-cost reparameterization.